



OPEN

Grain yellowness is an effective predictor of carotenoid content in global sorghum populations

Rae McDowell¹, Linly Banda¹, Scott R. Bean², Geoffrey P. Morris³ & Davina H. Rhodes¹✉

Identification of high carotenoid germplasm is crucial to assist breeders in provitamin-A biofortification of sorghum (*Sorghum bicolor* [L.] Moench). High-performance liquid chromatography is the gold standard for carotenoid quantification, however, it is not feasible for large scale phenotyping due to its high cost and low throughput. In this study, we tested the feasibility of using grain color as a high-throughput method of carotenoid biofortification breeding. We hypothesized that visual, color-based selection can be an effective strategy to identify high-carotenoid accessions. Yellow grain had significantly higher carotenoid content than red, brown, and white grain. The degree of yellowness could distinguish the presence or absence of carotenoids, but could not distinguish carotenoid concentrations within yellow-only accessions. The degree of luminosity of the grain, however, was able to better predict carotenoid concentrations within yellow-only accessions. Genome-wide association studies identified significant marker-trait associations for qualitative and quantitative grain color traits and carotenoid concentrations near carotenoid pathway genes—*ZEP*, *PDS*, *CYP97A*, *NCED*, *CCD*, and *LycE*—three of which were common between grain color and carotenoid traits. These findings suggest that using grain color as a method for screening germplasm may be an effective high-throughput selection tool for prebreeding and early-stage breeding in carotenoid biofortification.

Keywords Provitamin-A, Biofortification, Carotenoids, Cereals, Grain color, GWAS

Vitamin A deficiency (VAD) affects millions of people worldwide, contributing to decreased immune function and increased incidence of blindness and death in children under the age of five¹. The major source of vitamin A in global diets are carotenoids, which are yellow, red, and orange pigmented plant compounds. Carotenoids have a variety of functions in plants, including acting as photoprotectants, anti-oxidants, and hormone precursors. In humans, carotenoids contribute to protection against oxidation, cataracts, age-related macular degeneration, and chronic diseases². Provitamin A carotenoids such as β -carotene, β -cryptoxanthin, and α -carotene are compounds that are cleaved into vitamin A after they are consumed and are commonly produced in food plants. In many vitamin A deficient countries, vitamin A is obtained predominantly from provitamin A carotenoids in plant-based foods. Major staple cereal crops consumed in these regions, such as rice, wheat, maize, or sorghum, are significant contributors of calories, but are generally low in carotenoids.

Sorghum is a phenotypically and genetically diverse staple cereal crop grown for human food, animal feed, and fiber³. Some varieties of sorghum produce carotenoids in the grain, but the concentrations are low compared to fruits and vegetables^{4–7}. A 2009 study reported on 164 sorghum lines that have varying levels of carotenoid content, in which carotenoids in selected landraces ranged from: 0.23 to 0.66 $\mu\text{g/g}$ for lutein, 0.60 to 1.01 $\mu\text{g/g}$ for zeaxanthin, and 0.10 to 0.22 $\mu\text{g/g}$ for β -carotene⁵. In an assessment of global sorghum germplasm for carotenoid content, Cruet-Burgos et al. (2023) reported concentrations ranging from 0.02 to 4.61 $\mu\text{g/g}$ for lutein, 0.01–2.40 $\mu\text{g/g}$ for zeaxanthin, and 0.03–1.19 $\mu\text{g/g}$ for β -carotene. The provitamin α -carotene was detected in only 31 accessions, with values ranging from 0.02 to 0.11 $\mu\text{g/g}$. Identification or development of sorghum germplasm with higher concentrations of provitamin A carotenoids could contribute to efforts to alleviate VAD in sorghum-consuming regions. However, phenotyping carotenoid concentrations is expensive, time-consuming, and requires specialized knowledge of high-performance liquid chromatography (HPLC), the gold standard for carotenoid quantification. Therefore, a high-throughput phenotyping method could accelerate provitamin A carotenoid biofortification efforts.

¹Department of Horticulture and Landscape Architecture, Colorado State University, Fort Collins, CO 80523, USA. ²Center for Grain and Animal Health Research, USDA-ARS, 1515 College Ave, Manhattan, KS 66502, USA. ³Department of Soil and Crop Science, Colorado State University, Fort Collins, CO 80523, USA. ✉email: davina.rhodes@colostate.edu

Using grain color as a phenotypic selection method in carotenoid biofortification is potentially faster and cheaper than HPLC (Fig. 1a). There is some evidence that grain color is a predictor of carotenoid concentrations. For example, color has been shown to be a reliable marker for both carotenoid and polyphenol selection in maize, although these compounds generally accumulate in higher amounts in maize than sorghum^{8–10}. Sorghum grain can appear white, red, brown, or yellow depending on several factors, including the presence of polyphenol pigments in the pericarp, the thickness of the pericarp, and the concentration of yellow carotenoids in the endosperm^{11,12}. Sorghum endosperm color has been shown to be correlated with carotenoid concentrations, with the highest association found between endosperm color and total carotenoids, followed by zeaxanthin, and β -carotene ($R=0.71$, 0.69 , and 0.59 respectively)¹³. However, it is not known if grain color, rather than endosperm color, can be used as an even simpler approach to predict the concentration and types of carotenoids in the sorghum grain.

Genomic analyses suggest there is a relationship between grain color and carotenoid concentrations. In maize, studies using a visual color score found significant associations between kernel color and loci near the carotenoid pathway genes *ZEP*, *PSY*, *LycE*, and *CCD1*¹⁴, and a colorimetry study found significant associations between kernel color and loci near the carotenoid pathway genes *PSY*, *DXS2*, *LcyE*, and *ZEP*¹⁵. In a maize genome-wide association study (GWAS) for kernel color, several significant loci near genes in the carotenoid precursor (MEP) pathway and carotenoid pathway were identified¹⁶. These studies suggest that the use of grain color could be a proxy for carotenoid content in maize, and might similarly be used in sorghum.

In addition to visual selection, marker-assisted selection (MAS) could significantly decrease the time needed for biofortification of low-carotenoid staple cereals^{17,18}. Genetic markers can be used by breeders to select germplasm with preferred traits based on specific alleles at a loci, which can take less time and resources than phenotyping. MAS is particularly useful for grain-related traits, because favorable germplasm can be selected at the seedling stage rather than waiting until grain maturity to phenotype and select favorable germplasm. Molecular markers such as single nucleotide polymorphisms (SNPs) have been used to identify QTL and causal

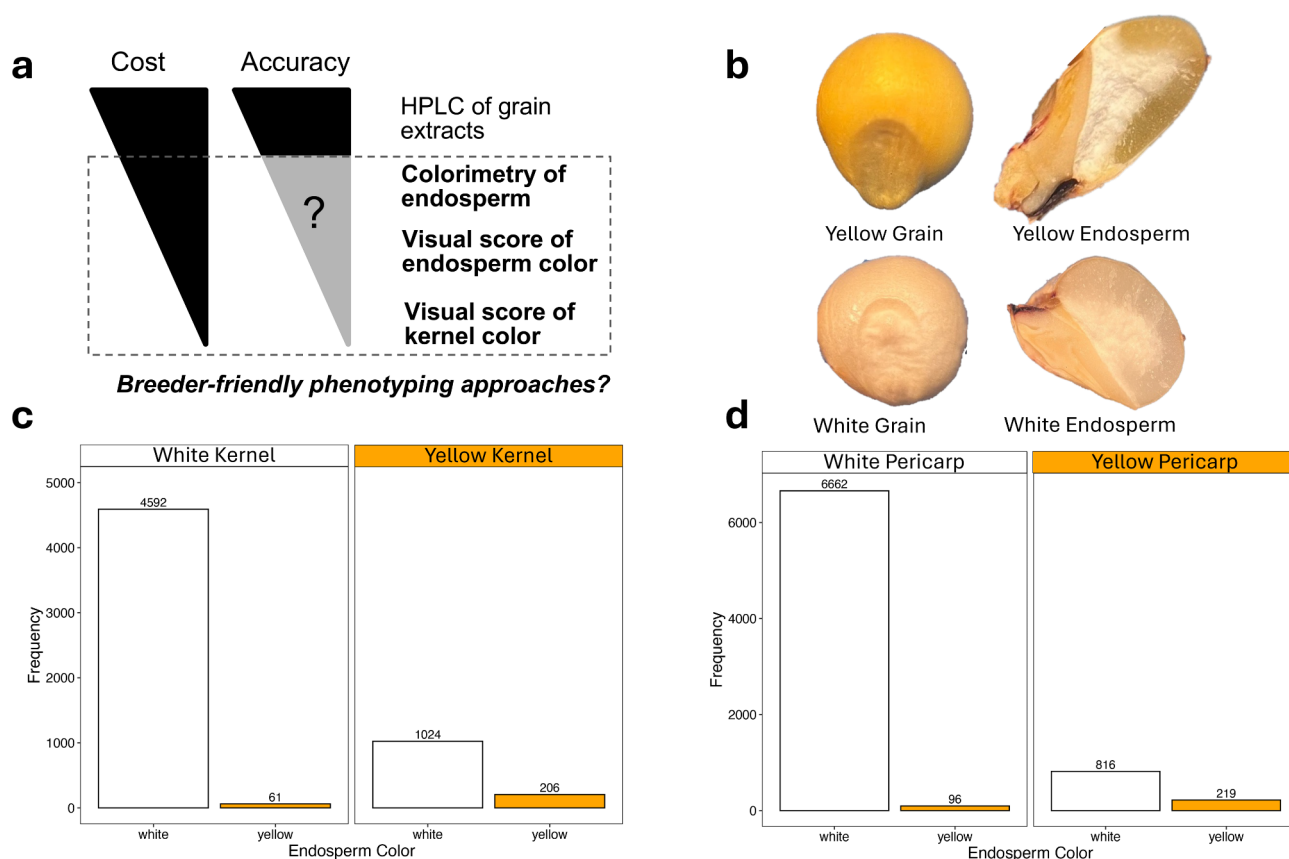


Fig. 1. Sorghum improvement programs need cost-effective approaches to identify high-carotenoid yellow-grain sorghum. **(A)** High-performance liquid chromatography is a highly accurate approach to quantify carotenoids, but it is too costly, in time and resources, for many breeding programs—particularly under-resourced programs in developing countries. Colorimetry, or even visual scoring, could provide more “breeder-friendly” phenotyping approaches, if the accuracy remains sufficiently high to capture genotypic differences relevant to pre-breeding and/or breeding programs. **(B)** Examples of yellow and white colored grain and endosperms. **(C)** Distribution of yellow endosperm between yellow and white grain accessions according to GRIN data for white and yellow kernel, and **(D)** white and yellow pericarp. Yellow endosperm is rare relative to white endosperm in global sorghum germplasm, and yellow pericarp may confound visual scoring.

variants, as well as to develop Kompetitive Allele Specific PCR (KASP) genotyping markers that can be used for early selection of favorable germplasm¹⁹. Given the associations between grain color and carotenoid pathway genes that have been identified in previous studies, it is possible that grain color KASP markers could be used for sorghum carotenoid biofortification.

The goals of this research were to (1) characterize the variation in carotenoid content of the yellow sorghum grain collection available in the USDA gene bank; (2) determine if grain color can be used to predict grain carotenoid concentrations; and (3) identify genetic markers associated with yellow grain, and determine if any co-localize with carotenoid pathway genes (Supplementary Fig. S1). We hypothesized that sorghum grain yellowness is an effective predictor for total carotenoid concentrations for selection in carotenoid biofortification in pre-breeding and breeding (Fig. 1a). Using both qualitative and quantitative assessment of grain color, we determined that grain yellowness could predict presence or absence of total carotenoids, but not carotenoid concentrations. However, between yellow accessions, the lightness or darkness of color could differentiate carotenoid concentrations to some degree. Finally, we identified several loci underlying variation in yellowness, three of which also contribute to variation in carotenoid concentrations.

Methods

Grain color analysis

Grain color phenotypes were obtained in three ways. First, we used data from the U.S. National Plant Germplasm System's Germplasm Resources Information Network (GRIN)²⁰. Data for endosperm color, kernel color, pericarp color, and testa pigmentation were used for analyses (Supplementary Data S1). GRIN defines pericarp color as "Color of the pericarp", and kernel color as "Phenotypic seed color (does not involve glumes)". It is unclear if reported values for these two categories clearly differentiate the color of the physical pericarp from the perceived overall color of the grain, which is influenced by testa and endosperm color, as well as pericarp thickness, so we used both traits in our analyses. For lines that were reported by GRIN to have both a pigmented testa and a yellow endosperm, validation was conducted by dissecting the grain and observing color under a dissecting microscope.

Next, we used a qualitative visual categorization of grain color using 323 accessions in the sorghum association mapping panel (SAP)²¹ and a subset of 81 accessions that we previously selected based on presence of yellow endosperm or yellow grain color (pericarp and kernel) according to data in GRIN⁷. The SAP contains accessions from all major morphological sorghum types (durras, caudatums, kafirs, and guineas) and centers of diversity in sub-Saharan Africa and Asia, as well as important breeding lines from the United States. The SAP has also been previously characterized^{7,22}, therefore we wanted to leverage available phenotype data to generate and test hypotheses. Three grains each per accession were scored independently by two individuals and classified as white, yellow, red, or brown (Supplementary Data S2).

Finally, we used a Konica Minolta (CR410) chroma meter colorimeter (Konica Minolta, U.S.A) to quantitatively measure the color of ground whole grain samples consisting of 185 accessions in the SAP (Supplementary Data S3). The colorimeter quantifies color differences using CIELAB color space: where L^* (luminosity) measures differences in lightness and darkness, a^* measures differences in red and green, and b^* measures differences in yellow and blue.

Carotenoid quantification

The panel was previously phenotyped using HPLC⁷. Briefly, using the 404 sorghum accessions described above, a single replicate of a complete randomized block design was planted in early June 2015 at Kansas State University's Department of Agronomy North Farm in Manhattan, Kansas. Panicles were collected at physiological maturity. Harvested panicles were oven dried, mechanically threshed, and then stored in a freezer until analysis. Carotenoid extraction and quantification have been previously described⁷. Briefly, grain samples were ground on a Udy Mill and stored in a -80°C freezer until ready for use. Extractions were conducted under yellow light to reduce degradation. Carotenoids were extracted with ethanol and saponified with KOH (80% w/v in H₂O). Solid phase extraction (SPE) cartridges with dichloromethane were used to clean up and concentrate the carotenoid extract. Extracts were then dried under nitrogen and resuspended in 25 μl MeOH:EtOAc (50:50 v/v). Two microliters of extract were injected into a YMC Carotenoid C30 column (3 μm , 2×150 mm, YMC American, Inc., Allentown, PA) for HPLC analysis and detected at 450 nm using a diode array detector on an Agilent 1260 HPLC (Agilent Technologies, Palo Alto, CA). Mobile phase A consisted of MeOH/H₂O (98:2 v/v), mobile phase B consisted of MeOH/H₂O (95:5 v/v), and mobile phase C consisted of MTBE. The carotenoid concentrations were reported in $\mu\text{g/g DM}$.

Genomic analysis

The panel was previously genotyped for 949,378 single nucleotide polymorphisms (SNPs)^{7,23}. SNPs with a minor allele frequency (MAF) less than 5% were filtered out in TASSEL 5.2.91²⁴, resulting in 348,181 SNPs remaining. GWAS were conducted for qualitative and quantitative (colorimetry) grain color, and for carotenoid concentrations. For each trait, only accessions with both phenotype and genotype data were used. Out of the available data in the study, 175 accessions were used for the yellow versus white grain GWAS, 262 accessions were used for the yellow versus not yellow GWAS, 185 accessions were used for the colorimeter GWAS, and 310 accessions were used for the carotenoid GWAS. There were 59 brown, red, or white accessions for which we had grain color data, but lacked GBS data, so had to be excluded from GWAS.

GWAS were performed using the Genome Association and Prediction Integrated Tool (GAPIT) 3.0²⁵ in R. The Bayesian Information and Linkage-disequilibrium Iteratively Nested Keyway (BLINK)²⁶ statistical method was used—which uses Bayesian information criteria in a fixed effect model—and the first seven principal components (PCA.total = 7) were included as covariates to control for population structure, reducing false positives. BLINK

was used for its high computing power and efficiency for running multiple traits, as well as for accounting for linkage disequilibrium (LD) in selection of significant markers. Significant marker-trait associations (MTAs) were identified using 5% false discovery rate (FDR) adjusted p-values to account for multiple comparisons. We conservatively considered carotenoid a priori candidate genes within 250 kb of significant MTAs, based on the reported LD decay of 175–600 kb in the SAP²⁷. Sorghum reference genome v3.1 from Phytozome was used²⁸. PCA plots were made using GAPIT PCA output data. Manhattan plots were made using GAPIT results with R packages “qqman” and “glmnet”. All analyses were conducted using R Studio version 4.2.1²⁹.

Results

Characterization of genebank germplasm to identify new sources of high carotenoid sorghum

We first sought to characterize the yellow sorghum germplasm that is available through the USDA gene bank. Criteria used to select accessions was presence of (1) yellow endosperm, (2) yellow pericarp, or (3) yellow kernel, according to data reported in GRIN. Of the 14,537 accessions with endosperm color data, less than 3% (379 accessions) were characterized as having a yellow endosperm, and another 2% (291) were characterized as having a mixed (yellow and white) endosperm (Supplementary Fig. S2). Of the 14,360 accessions with pericarp color data, 7% (1070 accessions) were characterized as having a yellow pericarp, and another 10 accessions were characterized as having a mixed (yellow and other) pericarp (Supplementary Fig. S2). Finally, of the 18,742 accessions with kernel color data, 12% (2284 accessions) were characterized as having a yellow kernel, and another 25 accessions were characterized as having a mixed (yellow and other) kernel color (Supplementary Fig. S2).

Next, we hypothesized that yellow grain color (either yellow pericarp or yellow kernel) corresponds with a yellow endosperm. There were 5883 accessions in GRIN with both endosperm and kernel color data, and 7793 accessions with both endosperm and pericarp color data, with 5407 accessions in common between the two datasets. Since flavonoid compounds in brown and red grain can mask yellowness, we only included accessions listed as white or yellow grain and endosperm (Fig. 1b). Of the 4653 accessions listed as having a white kernel, only 61 of the accessions (1%) were listed as having a yellow endosperm (Fig. 1c). In contrast, of the 1230 accessions listed as having a yellow kernel, 206 of the accessions (17%) were listed as having a yellow endosperm (Fig. 1c). Similarly, of the 6758 accessions listed as having a white pericarp, only 96 of the accessions (1%) were listed as having a yellow endosperm, and of the 1035 accessions listed as having a yellow pericarp, 219 of the accessions (21%) were listed as having a yellow endosperm (Fig. 1d).

Next, we categorized the geographic distribution of yellow sorghum accessions based on GRIN data. Geographic origins of yellow sorghum accessions can help to identify regions that may be sources of new high carotenoid sorghums. To determine if there were any patterns of geographic distribution, we aggregated 327 yellow endosperm accessions (Fig. 2a), 799 yellow pericarp accessions (Fig. 2b), and 1604 yellow kernel accessions (Fig. 2c).

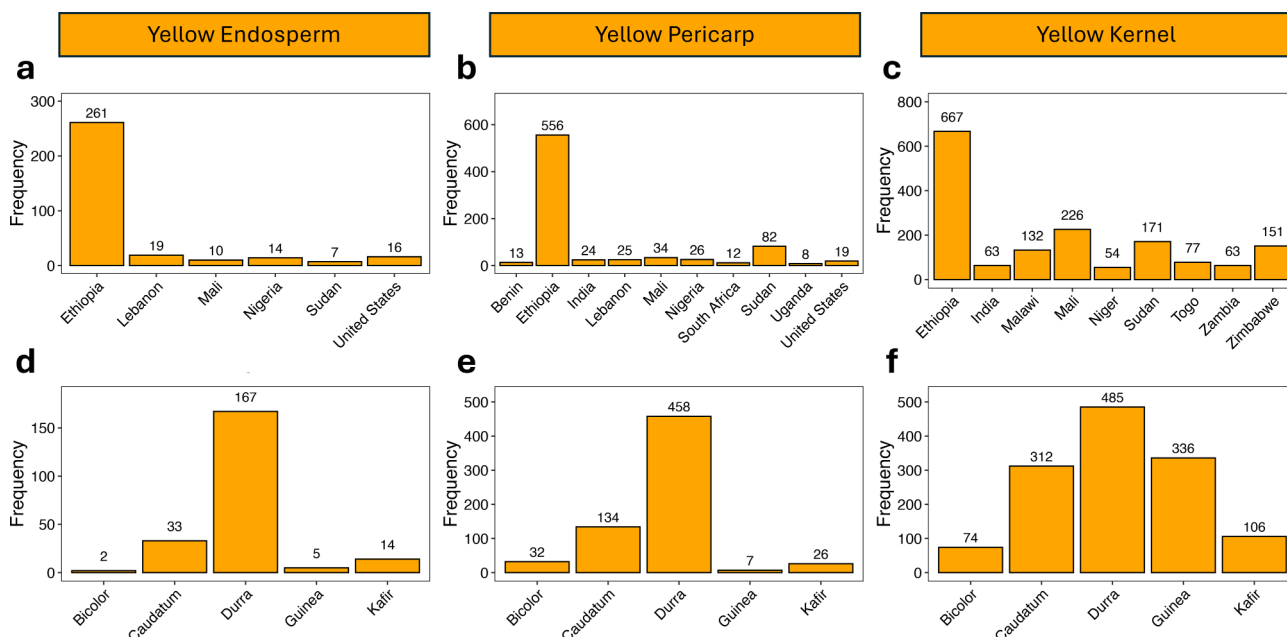


Fig. 2. Most yellow grain sorghum is from Ethiopia and of the durra botanical type. Distribution of yellow sorghum accessions by geographic origin of (A) yellow endosperm, (B) yellow pericarp, (C) yellow kernel; and morphological type of (D) yellow endosperm, (E) yellow pericarp, and (F) yellow kernel. Mixed morphological types were not included. Countries were included in the analysis if the number of yellow endosperm was > 5, yellow pericarp > 5, and yellow kernel > 50.

accessions (Fig. 2c) by country of origin. The vast majority of all three categories originated in Ethiopia. Many yellow endosperm accessions also originated from Lebanon, United States, and Nigeria.

To further identify potential sources of yellow sorghum accessions, we investigated patterns of distribution of 221 yellow endosperm (Fig. 2d), 657 yellow pericarp (Fig. 2e), and 1313 yellow kernel (Fig. 2f) accessions among sorghum morphological types. The majority of accessions from all three categories were classified as durras: yellow endosperm (76%), yellow pericarp (70%), yellow kernel (37%). Given similar distribution patterns of yellow kernel and yellow pericarp types (Fig. 1c and d, Fig. 2), in subsequent analyses we combined these to represent generalized yellow grain color.

We also investigated the relationship between yellow endosperm sorghum and condensed tannins, which are pigmented compounds produced in the pericarp and testa of some sorghum varieties and is an important trait in breeding selection, based on end-user preferences. The presence of both condensed tannins and carotenoids in the grain can influence overall color perception, thereby complicating the usefulness of visual selection for carotenoids. Presence of condensed tannins can be visually confirmed by the presence of a pigmented testa. To identify carotenoid producing sorghums that also produce condensed tannins, we aggregated endosperm color by presence or absence of a pigmented testa, using GRIN data (Supplementary Fig. S3). Of the 12,394 accessions with a white endosperm, 5742 (46%) had a pigmented testa and 6652 (54%) did not. Notably, of the 370 accessions with a yellow endosperm, 19 (5%) had a pigmented testa and 351 (95%) did not.

To determine if presence of a pigmented testa interferes with visual selection of a yellow endosperm, the 19 accessions listed in GRIN as having both a yellow endosperm and a pigmented testa were ordered from GRIN and phenotyped by cutting the grain in half and observing under a microscope. We first validated the GRIN phenotypic scores of endosperm color and testa presence, particularly since according to GRIN, the majority of yellow endosperm accessions do not have a pigmented testa. Of the 19 accessions with a reported pigmented testa, we observed that the pigmented testa was present in only 6, segregating in 4, and absent in 9 (Supplementary Fig. S4). Furthermore, of the 19 accessions with a reported yellow endosperm, only 7 were yellow, 3 were white, and the remaining were mixed or the color was masked by the testa (Supplementary Fig. S5). The presence of a pigmented testa seemed to block light into the endosperm and create the appearance of a reddish brown hue in the endosperm (Supplementary Fig. S6). The accurate assessment of endosperm color was only possible when the testa was removed by scraping it away with a blade (Supplementary Fig. S6). Notably, we observed no accessions with both a yellow endosperm and a pigmented testa.

Measurements of yellowness predict presence or absence of carotenoids

Next we wanted to ascertain if grain color differences could predict carotenoid content. Overall, the most abundant carotenoids were lutein and zeaxanthin (1.4 and 0.8 $\mu\text{g/g}$, respectively). Amongst the provitamin A carotenoids, only β -carotene was detected at (0.09 $\mu\text{g/g}$), while β -cryptoxanthin and α -carotene were below the limit of detection in the majority of the samples and were thus excluded from further analysis. To determine if there were differences in carotenoid concentration based on a qualitative categorization of grain color (yellow, white, red, brown), we conducted ANOVAs between carotenoid concentration and grain color in 321 sorghum accessions. Yellow grain sorghum accessions had significantly higher concentrations of lutein ($P < 10^{-15}$) (Fig. 3a), zeaxanthin ($P < 10^{-15}$) (Fig. 3b), β -carotene ($P < 10^{-15}$) (Fig. 3c), and total carotenoids ($P < 10^{-15}$) (Fig. 3d, and Supplementary Table S1) than red, brown, or white grain accessions, whereas there were no significant differences in carotenoid concentrations between non yellow accessions.

Among yellow-only accessions, visual assessment could not easily differentiate grain yellowness, so we questioned if a quantitative measurement of grain color could be used to identify yellow accessions with the highest carotenoid concentrations. A colorimeter was used to quantitatively measure yellowness of whole ground grain. We used ground samples for this, as it was more effective in NIRS analysis compared to whole grain. We first measured grain color differences in accessions of all color-categories, and Pearson's correlations were conducted between colorimeter values and carotenoid concentrations (Table 1a). There were significant positive correlations between b^* values (degree of yellowness) and total carotenoids ($r = 0.40$, $P = < 10^{-7}$), β -carotene ($r = 0.36$, $P = < 10^{-6}$), zeaxanthin ($r = 0.38$, $P = < 10^{-6}$), and lutein ($r = 0.40$, $P = < 10^{-7}$) concentrations. However, the significant correlation disappeared when comparing the b^* values of only yellow grain accessions (Table 1b), indicating that quantitative measurements of yellowness of the whole ground grain was no better than a visual qualitative assessment for selecting carotenoid-containing accessions.

There were significant negative correlations between L^* values (grain lightness) and total carotenoids ($r = -0.20$, $P = 0.006$), β -carotene ($r = -0.19$, $P = 0.009$), zeaxanthin ($r = -0.16$, $P = 0.03$), and lutein ($r = -0.22$, $P = 0.002$) concentrations in the entire panel (Table 1a), and interestingly, the negative correlation increased when conducting the analysis on only yellow grain accessions for total carotenoids ($r = -0.51$, $P = 0.004$), β -carotene ($r = -0.50$, $P = 0.005$), zeaxanthin ($r = -0.42$, $P = 0.02$), and lutein ($r = -0.53$, $P = 0.003$) concentrations (Table 1b). These results suggest that grain lightness could potentially be used as a high-throughput method to identify accessions with higher carotenoid concentrations, but further experiments in additional populations are needed to validate this approach.

GWAS identifies loci underlying variation in yellowness near carotenoid a priori candidates

Since yellow grain sorghum accessions have significantly higher carotenoid concentrations than brown, red, or white grain sorghum, we sought to identify MTAs that could potentially be used in MAS for carotenoid biofortification. We also sought to determine if there is overlap in genes controlling grain color and grain carotenoid concentrations. First, to determine the amount of genetic diversity within the yellow sorghum accessions, a principal component analysis (PCA) was conducted using the genotype data, highlighting grain color differences (Fig. 4). In a PCA for both the first and second (Fig. 4a) and first and third principal components (Fig. 4b), yellow grain sorghum clustered together, indicating the yellow grain sorghums in the

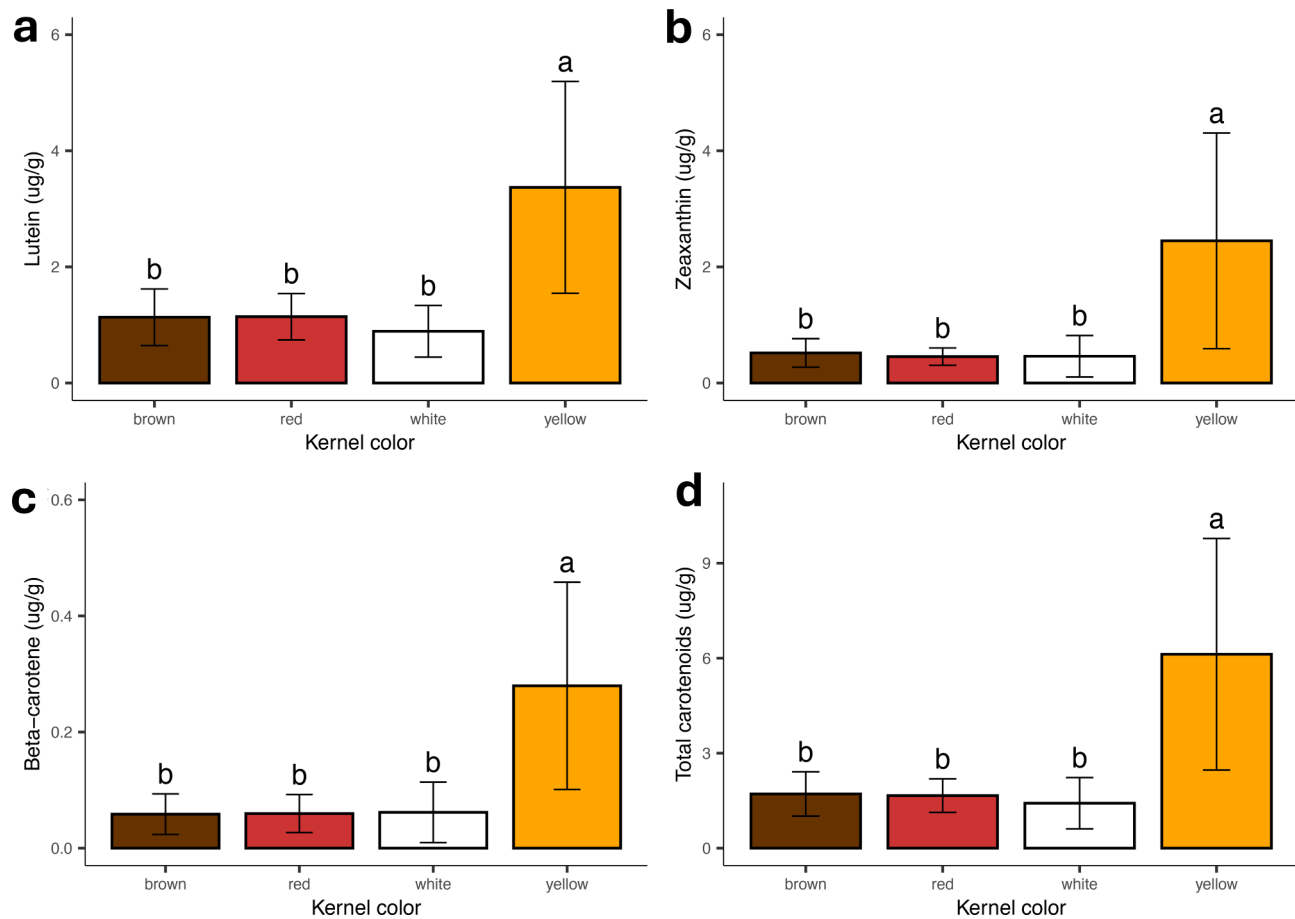


Fig. 3. Sorghum accessions classified visually as “yellow” have higher provitamin A and non-provitamin A carotenoids. Mean carotenoid concentrations between visual grain color classifications for (A) lutein, (B) zeaxanthin, (C) β -carotene, and (D) total carotenoids in accessions of each grain color. Color categories with different letters have significantly different carotenoid concentrations from each other, based on a post hoc Tukey HSD test, with significant P -value < 0.05 (brown, $n = 89$; red, $n = 27$; white, $n = 156$; yellow, $n = 49$).

	a*	b*	Lutein	Zeaxanthin	β -carotene	Total carotenoids
(a) All grain colors						
L	−0.83***	0.23**	−0.22**	−0.16*	−0.19**	−0.20**
a*		−0.38***	−0.09	−0.14	−0.11	−0.12
b*			0.40***	0.38***	0.36***	0.40***
Lutein				0.86***	0.94***	0.97***
Zeaxanthin					0.85***	0.96***
β -carotene						0.94***
(b) Yellow grain only						
L	−0.72***	0.42*	−0.53**	−0.42*	−0.50**	−0.51**
a*		−0.57***	0.20	0.12	0.18	0.17
b*			0.17	0.16	0.09	0.18
Lutein				0.74***	0.89***	0.93***
Zeaxanthin					0.69***	0.92***
β -carotene						0.86***

Table 1. Relationship of grain color and carotenoids. Pearson’s correlations among mean colorimetry values and carotenoid concentrations in (a) accessions of all grain colors ($n = 185$), and (b) accessions of yellow grain only ($n = 30$). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

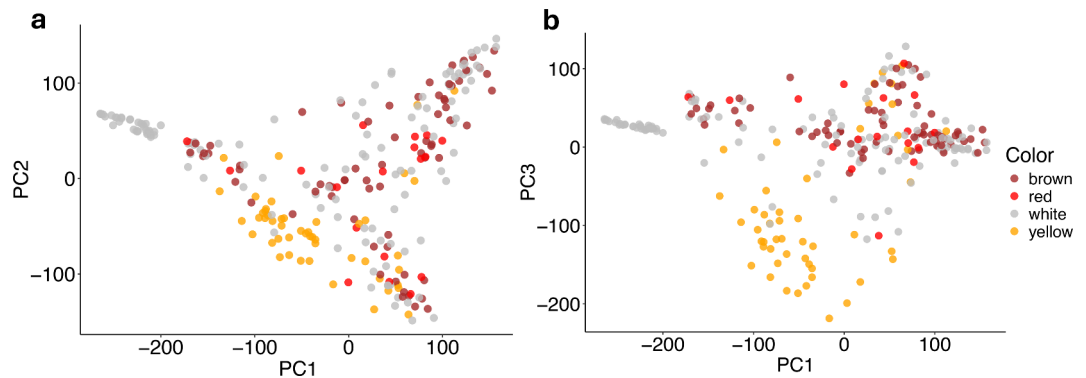


Fig. 4. Grain color is structured by subpopulation in global sorghum diversity. Accessions plotted according to (A) the first two principal components, and (B) the first and third principal components for sorghum kinship, and coded by color of sorghum grain: brown ($n=69$), red ($n=18$), white ($n=126$), yellow ($n=49$).

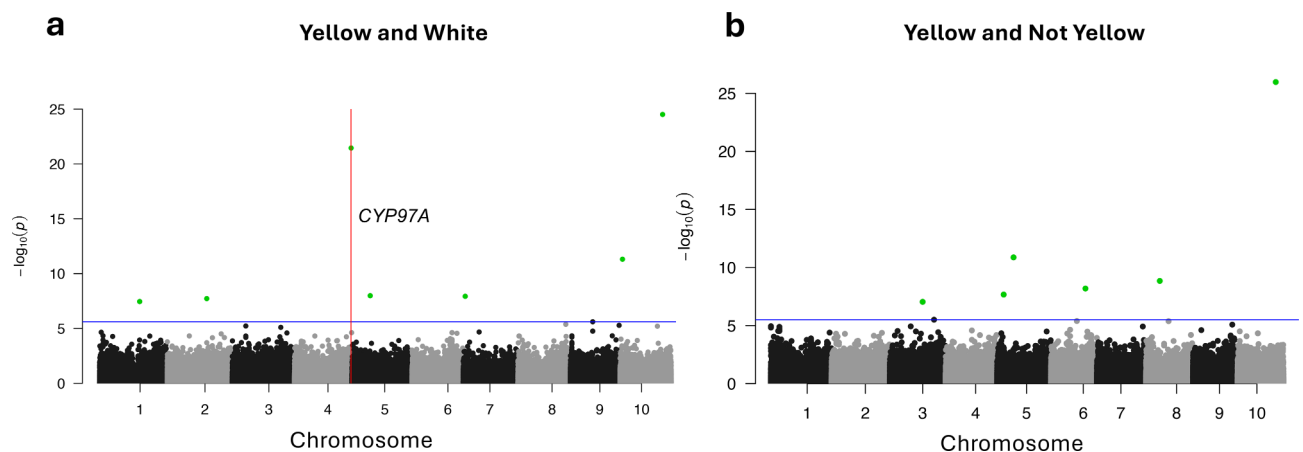


Fig. 5. GWAS identifies candidate loci for variation of qualitative grain color in global sorghum germplasm. Manhattan plot of association for (A) yellow ($n=49$) and white ($n=126$) grain color, and (B) yellow ($n=49$) and not yellow ($n=213$) grain color. The horizontal blue line represents the genome-wide significance threshold at 5% FDR. The vertical red line represents the start site of Sobic.004G346000 (*CYP97A*).

panel are genetically similar. In contrast, red, white, and brown sorghum did not cluster as tightly by color, indicating greater genetic diversity than the yellow grain sorghums.

MTAs for qualitative grain color

To identify regions underlying qualitative grain color, we conducted a GWAS using grain color phenotypes that were visually identified as white, yellow, red, or brown. We first used only yellow and white grain accessions (Fig. 5a). Brown and red grain have high concentrations of flavonoid compounds in the pericarp¹², so were excluded from analysis to reduce confounding factors. GWAS identified seven significant MTAs (Supplementary Table S2), one of which (S4_67817974; $P < 10^{-8}$) was 237 kb away from *CYP97A*, a carotene β -ring hydroxylase (also known as *LUT5*; Sobic.004G346000) on chromosome 4. Notably, just below the 5% FDR threshold, there was a small peak on chromosome 6 (47.2–47.9 Mb), in proximity to the a priori gene candidate *zeaxanthin epoxidase* (*ZEP*; Sobic.006G097500) (Fig. 5a and Supplementary Fig. S7 and Table S2), a major gene underlying sorghum carotenoid variation^{7,22}.

To determine if any additional grain color MTAs could be identified that colocalized with carotenoid MTAs, we conducted a GWAS using all grain colors, and classified each accession as either yellow or not yellow (Fig. 5b). GWAS identified six significant MTAs at 5% FDR (Supplementary Table S3), none of which were in proximity to candidate genes. One MTA (S10_50918381; $P < 10^{-26}$) was detected by both GWAS. One MTA in the yellow or not yellow GWAS was upstream of *ZEP* on chromosome 6 at 45.98 Mb however this was beyond our 250 kb limit.

MTAs for quantitative (colorimetry) grain color

To identify regions underlying quantitative grain color, we conducted a GWAS using colorimeter ($L^*a^*b^*$ values) values. Again, we first conducted the GWAS with only yellow and white grain accessions to reduce confounding

factors. GWAS did not identify any significant MTAs for either L^* , a^* or b^* values, likely due to the low sample size ($n=64$), however, several MTAs were near the significance threshold (Supplementary Fig. S8).

Next, we conducted a colorimetry GWAS using accessions with all grain colors to try to identify significant MTAs for any color (Fig. 6 and Supplementary Table S4). GWAS identified four significant MTAs for L values (luminosity axis), one of which (S6_53091871; $P < 10^{-8}$) was 137 kb away from a putative phytoene desaturase (PDS: Sobic.006G177400), a a priori candidate gene for carotenoid biosynthesis in sorghum (Fig. 6a). GWAS also identified four significant MTAs for a^* values (red vs. green axis), none of which were near carotenoid a priori candidate genes (Fig. 6b). GWAS did not identify significant MTAs for b^* values (yellow vs. blue axis) (Fig. 6c).

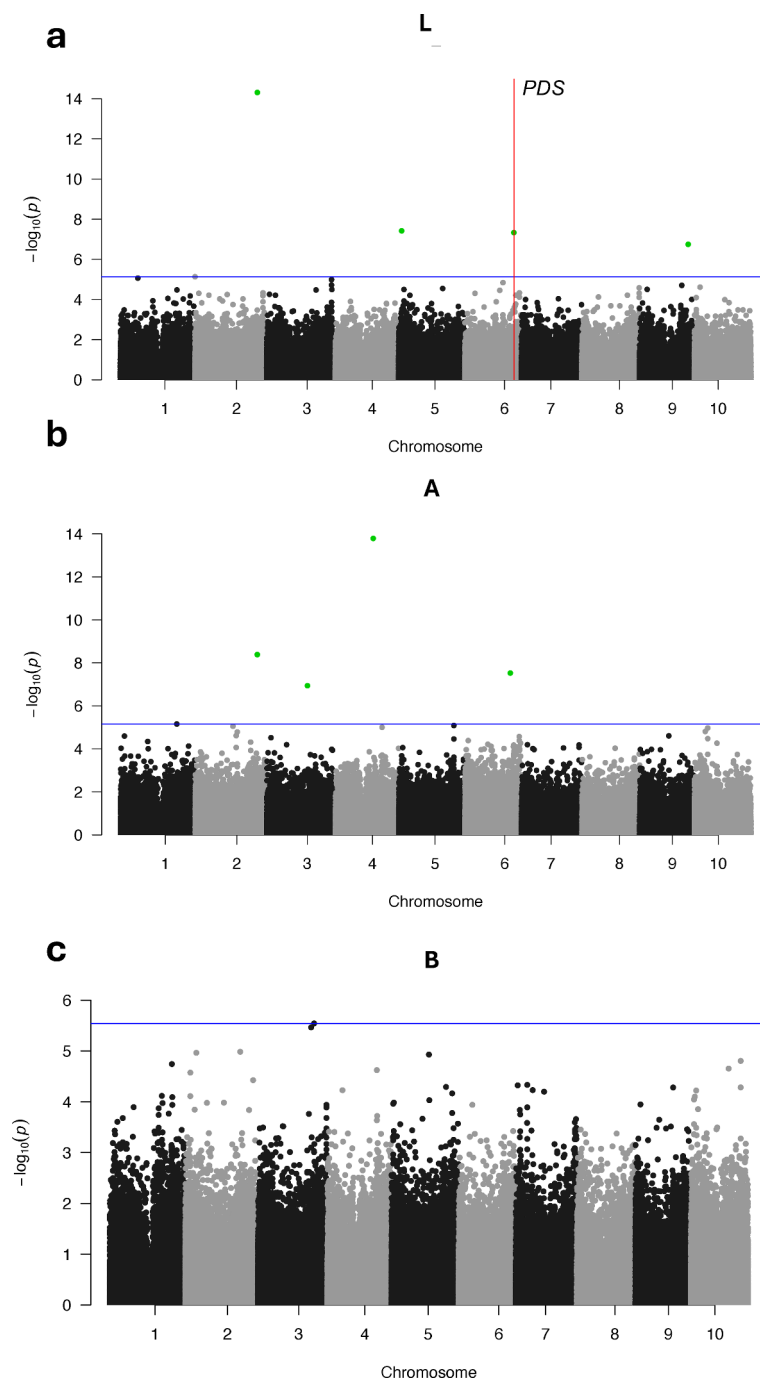


Fig. 6. GWAS on colorimetric values identify additional candidate loci for grain color variation ($n=185$). Manhattan plot of association results for (A) L^* values, (B) a^* values, and (C) b^* values. The horizontal blue line represents the genome-wide significance threshold at 5% FDR. The vertical red line represents the start site of Sobic.006G177400 (PDS).

Finally, we hypothesized that there are common genetic controls for variation in carotenoid concentrations and grain color, so we conducted a GWAS for lutein, zeaxanthin, β -carotene, and total carotenoids. While this carotenoid concentration data has been analyzed in our previous study⁷, we sought a direct comparison of carotenoid GWAS versus color GWAS using the same statistical model. The total carotenoid GWAS identified 10 significant MTAs, one of which was within *ZEP*, a candidate gene on chromosome 6 (S6_46717975; $P < 10^{-11}$) (Fig. 7a and Supplementary Table S5). The lutein GWAS identified 28 significant MTAs, two of which were near carotenoid candidate genes (Fig. 7b and Supplementary Table S5). An MTA on chromosome 3 (S3_52257288; $P < 10^{-9}$) was just 12 kb from a putative *lycopene epsilon cyclase* gene (*LycE*; Sobic.003G197400). Another MTA on chromosome 4 (S4_61450172; $P < 10^{-9}$) was 180 kb away from a putative *9-cis-epoxycarotenoid dioxygenase* gene (*NCED*; Sobic.004G268500).

The β -carotene GWAS identified one significant MTA, which was not near any carotenoid candidate genes (Fig. 7c and Supplementary Table S5). At 10% FDR, three more significant MTAs were identified—one of which (S2_60899695; $P < 10^{-11}$) was also identified in the GWAS for lutein. The zeaxanthin GWAS identified 19 significant MTAs, two of which were near carotenoid candidate genes (Fig. 7d and Supplementary Table S5). As with the total carotenoid GWAS and the yellow or not yellow GWAS, a significant MTA on chromosome 6 (S6_46717975; $P < 10^{-11}$) within the *ZEP* gene was identified. Another significant MTA on chromosome 7 (S7_60753862; $P < 10^{-7}$) was 248 kb from a putative *carlactone synthase* gene (*CCD8*; Sobic.007G170300).

Discussion

Utility of genebanks as a resource for identifying high carotenoid germplasm

Genebanks are a valuable resource for germplasm access and trait exploration³⁰. Through the GRIN database, we were able to identify and obtain hundreds of global sorghum accessions with potentially high carotenoid concentrations in the USDA genebank^{7,31}. Through characterization of the grain traits using data from GRIN and data generated in our lab using seeds from GRIN, we have identified germplasm that will be used for carotenoid biofortification breeding efforts, identified geographic regions in which to search for additional high carotenoid germplasm, and characterized some of the underlying genetic controls of carotenoid variation^{7,22,32}. While we focused here on biofortification value, carotenoids (and their apo-carotenoid derivatives) are also of interest for abiotic stress (e.g. xanthophyll cycle and abscisic in drought tolerance) and biotic stress (e.g. strigolactones in *Striga* resistance)³³.

Due to the subjective nature of visually phenotyping grain color, compounded by genotype by environment interactions that can change the color appearance from year to year and by location, the GRIN descriptor data should be validated when possible, and analyses with the data need to be interpreted cautiously. Still, previous studies have shown that genebank descriptor data can be directly usable for GWAS²³. We validated GRIN data

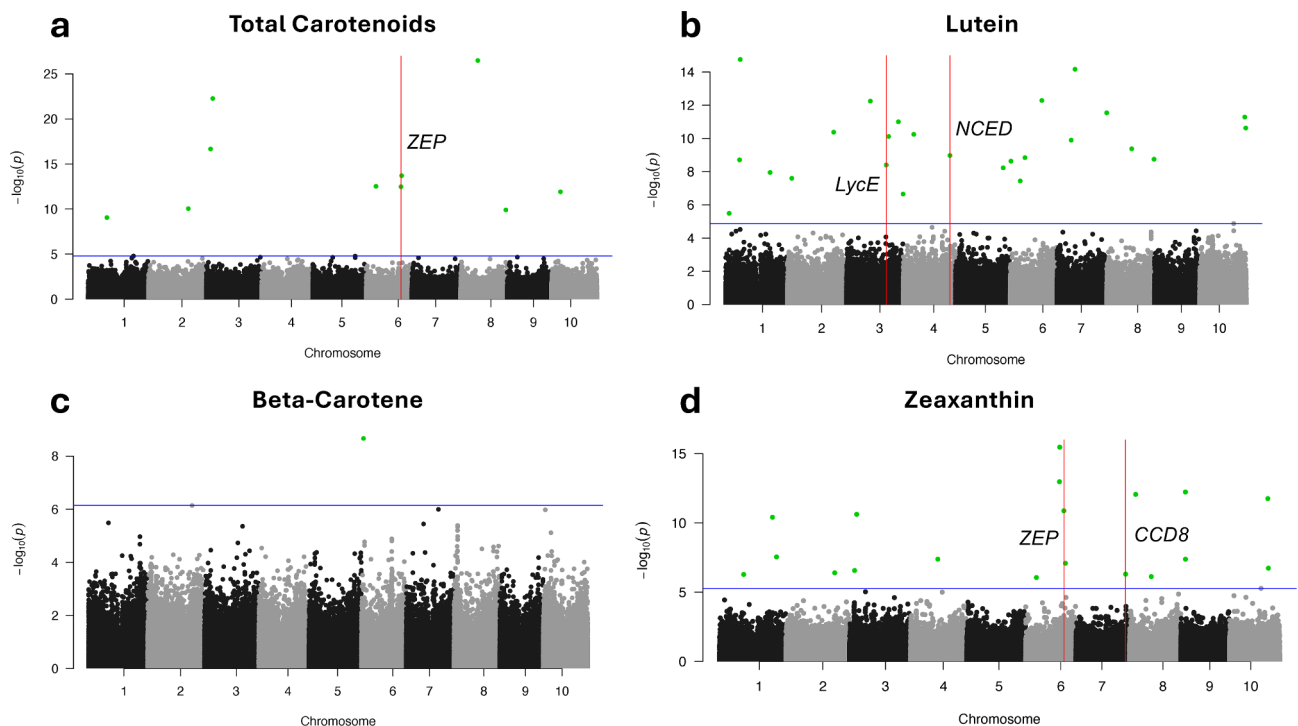


Fig. 7. GWAS of grain carotenoid concentrations in global sorghum germplasm reveals additional candidate loci. Manhattan plot of association results ($n = 310$) for (A) total carotenoids, (B) lutein, (C) β -carotene, and (D) zeaxanthin measured by HPLC. The horizontal blue line represents the genome-wide significance threshold at 5% FDR. The vertical red lines represent the start sites of the labeled candidate genes: Sobic.006G097500 (*ZEP*), Sobic.007G170300 (*CCD8*), Sobic.003G197400 (*LycE*), Sobic.004G268500 (*NCED*).

by evaluating the 19 accessions that were reported in GRIN as having both a pigmented testa and a yellow endosperm (see Supplementary Fig. S3 and S4). Although both carotenoids and polyphenols are known to be upregulated during stress and oxidation events in plants³⁴, we have previously observed that none of our high carotenoid accessions have a pigmented testa. One possible explanation for this is that there may be flux in the pathways that shunts exclusively towards carotenoid or flavonoid synthesis^{35,36}.

Our evaluation using a dissecting microscope did not identify any accessions with both a pigmented testa and a yellow endosperm. Some of the accessions did not have a pigmented testa, some did not have a yellow endosperm, and some were segregating for the traits (see Supplementary Fig. S3 and S4). Precise phenotyping of grain traits can be surprisingly challenging, particularly due to the subjective nature of color, so variation in observations is expected^{37–39}. For example, the presence of a testa complicated the evaluation of endosperm color by creating the appearance of a reddish brown hue in the endosperm, making it difficult to accurately discern the endosperm color (see Supplementary Fig. S5). Similarly, the presence of a large yellow embryo might be mistaken for yellowness in the endosperm. Altogether, our validation highlights the limitations of the GRIN data, but our analyses demonstrate that the descriptor data is a useful starting point for testing hypotheses and finding accessions with potentially high carotenoid concentrations.

Strengths and limitations of grain yellowness as a predictor of carotenoid concentrations

High-throughput selection methods (Fig. 1a) could reduce the number of breedings cycles needed to develop biofortified sorghum. Significantly higher carotenoid concentrations in the yellow sorghum grain versus white, red, or brown grain indicates that a qualitative visual assessment of grain color can be used as a general predictor of carotenoid concentrations in yellow versus non-yellow sorghum (Fig. 3). One limitation of using grain color is the lack of standardized definition across databases such as GRIN, where it is unclear if they are considering the color of the kernel, pericarp, or generalized color of the seed. Carotenoids such as β -carotene, β -cryptoxanthin, and α -carotene are generally orange-colored, whereas lutein and zeaxanthin are generally yellow, but given the very low concentrations of the carotenes in sorghum compared to lutein and zeaxanthin, the high carotenoid grain primarily appears yellow. The high carotenoid concentrations in yellow grain sorghum suggests that many of the varieties appear yellow either because the yellow grains have a thin translucent pericarp that allows the yellow endosperm to be seen through it, and/or because carotenoids also accumulate in the pericarp, contributing to the appearance of yellow grain⁴⁰. Some monocot seeds, including wheat, barley, and sorghum, have photosynthetic capability in the pericarp, which requires the presence of carotenoids^{41,42}. Quantifying carotenoid concentrations in individual grain tissues, and determining if the genetic controls are independent, could provide insights into ways to increase total carotenoids in the grain. The significant correlation between total carotenoids and provitamin A carotenoids such as β -carotene (Table 1a and b) supports the hypothesis that selection for total carotenoids can be a strategy to indirectly increase provitamin A carotenoids in sorghum. It can be difficult to visually quantify the variation in yellowness accurately¹⁴, so we used colorimetry to quantitatively measure differences in yellowness. There were significant correlations between degree of yellowness and concentrations of each carotenoid compound when analyzing all accessions, however, the significant correlations disappeared when analyzing the subset of yellow-only accessions (compare Table 1a. vs. b). This suggests that variation in grain yellowness in the diversity panel used in this study is more useful in identifying the presence or absence of carotenoids, but may not be used as a proxy for carotenoid concentrations. This is consistent with a maize study in which there were poor correlations between visually scored grain yellowness and carotenoid content in a panel of yellow or orange-only diverse inbreds⁴³. Additionally, previous studies⁴⁴ found that maize kernel hardness influenced the perception of kernel yellowness, further complicating accurate color phenotyping. Another consideration in phenotyping grain color is the presence of additional pigmented compounds in the pericarp, such as anthocyanins, condensed tannins, and flavanones, that potentially mask the yellowness from carotenoid compounds.

Potential for trait-predictive marker technology in carotenoid biofortification

The discovery of common genetic regulation between grain yellowness and carotenoid concentrations is further evidence to support our hypothesis that grain yellowness can be used as a proxy for carotenoid content. Although a single replication may not capture the phenotypic variation due to the environment, grain color is highly heritable—for example, brown grain due to proanthocyanidins^{12,45}—so we are confident that genomic mapping for yellow grain can identify meaningful marker-trait associations. Additionally, identifying loci underlying grain yellowness potentially provides us with a better understanding of carotenoid genetic controls, as well as additional markers to test for use in MAS. GWAS identified MTAs within and near *ZEP* for qualitative grain color (just below the 5% FDR threshold), as well as for total carotenoids and zeaxanthin (Fig. 5, 7a,d). *ZEP* has been identified in our previous studies^{7,22}, as well as in several other species, as a major controller of carotenoid variation^{16,46}. GWAS for quantitative color variation (colorimetry) identified an MTA near a *PDS* gene (Sobic.006G177400) that we previously identified in a transcriptomics study, in which the *PDS* expression was significantly correlated with concentrations of zeaxanthin in mature sorghum grain³². Additionally, the GWAS for β -carotene and yellow-white color both identified a common MTA on chromosome 10 (S10_2941924), although it was not near any a priori candidates. These results support the hypothesis that there is common genetic regulation between grain color and carotenoid variation, but further testing, such as biparental mapping, is needed for validation.

The GWAS for lutein variation identified an MTA very near a *LycE* gene (Sobic.003G197400) (Fig. 7b). Our previously published GWAS study for variation in the ratio of lutein and zeaxanthin also identified an MTA near this *LycE* gene⁷. *LycE* and *LycB* catalyze the cyclization of lycopene at the primary branch point of the carotenoid biosynthesis pathway (Supplementary Fig. S1), and therefore regulate the proportion of carotenes directed to each branch. In maize and wheat, *LycE* alleles with low enzyme activity have been identified, and they correspond

with an increased flux towards the β - β branch and higher β -carotene concentrations^{43,47}. The variation in both *CYP97A* and *LycE* has been successfully used in MAS for provitamin A in maize and wheat^{48–50}, and may be good candidates to test for sorghum MAS.

GWAS also identified MTAs near carotenoid candidate genes that we have not previously identified: *CYP97A* (Fig. 5; qualitative color GWAS), *NCED* (Fig. 7b; lutein GWAS), and *CCD8* (Fig. 7d; zeaxanthin GWAS). *CYP97A* is responsible for the β -ring hydroxylation of α - and β -carotene, with α -carotene being the most preferred substrate. The *CYP97A* identified in this study (Sobic.004G346000) is the ortholog to the maize *CYP97A* gene (Zm00001eb258960), so is a promising candidate to test in MAS.

CCDs and NCEDs are the main degradation enzymes that oxidatively cleave carotenoids to generate apocarotenoids, which contribute to color and scent of fruits and flowers, as well as to the production of two key plant hormones, strigolactone and abscisic acid³³. Downregulation of CCDs has been associated with an increase in total carotenoids, so is of interest for biofortification efforts⁵¹. Interestingly, the CCD identified as a candidate gene in this study (Sobic.007G170300) (Fig. 7d) was not found to be expressed in our previous grain expression study, but was identified in a sorghum root expression study^{32,52}. In contrast, the NCED (Sobic.004G268500) identified in the lutein GWAS was previously found to be differentially expressed between high and low carotenoid accessions in our expression study, although overall expression was relatively low.

The combined results of grain color and carotenoid GWAS identified several significant regions of association, but only six were near carotenoid a priori candidates (Supplementary Table S6). These results are consistent with our previous GWAS studies in which the majority of MTAs were not near known carotenoid genes⁵³, suggesting that there are genes involved in yellow pigmentation and carotenoid concentration that have yet to be characterized. It is possible that GWAS was unable to detect significant MTAs near other candidate genes due to reduced power from the small sample size of yellow grain accessions. Many of the yellow grain accessions are photoperiod sensitive and do not easily produce seed in the temperate regions where our samples were grown, so we were limited in the number of accessions we could grow. Increasing the number of accessions with both GBS and phenotype data would help increase mapping power to identify additional MTAs. To validate our current results and identify casual allelic variants, QTL analysis, sequence analysis, and marker testing could be conducted.

Conclusion

Our results emphasize the difficulty in grain color phenotyping, but suggest that grain color can be used in selections for high carotenoid germplasm during prebreeding and during early selection cycles of biofortification breeding (Fig. 3, Table 1a). In later stages of breeding, however, more accurate selection methods are needed that are better able to differentiate carotenoid concentrations within carotenoid-containing yellow grain germplasm (Table 1b). Development and testing of the common MTAs found between grain color and carotenoid concentrations is underway and could provide trait-predictive KASP markers that can help speed up the breeding process, as well as help to maintain the desirable carotenoid alleles in a breeding program's germplasm (Figs. 4, 5, 6, 7). Altogether, these results suggest that high-throughput grain color screening in prebreeding and early breeding stages, followed by MAS in later breeding stages, could be an effective strategy for successful sorghum carotenoid biofortification breeding.

Data availability

Original publication with genotype data²³. Original publication with carotenoid concentration data⁷. All other data files are available in the supplemental files and code is available at: <https://github.com/rmcdowder/sorghumbiofortification/tree/a8457f87068867eb687235c255a6222863102e1a>. Data can be found in the above publications or website, or requests can be sent to corresponding author Rhodes.

Received: 30 March 2024; Accepted: 4 October 2024

Published online: 24 October 2024

References

1. Sherwin, J. C., Reacher, M. H., Dean, W. H. & Ngondi, J. Epidemiology of vitamin A deficiency and xerophthalmia in at-risk populations. *Trans. R. Soc. Trop. Med. Hyg.* **106**, 205–214 (2012).
2. Eggersdorfer, M. & Wyss, A. Carotenoids in human nutrition and health. *Arch. Biochem. Biophys.* **652**, 18–26 (2018).
3. Monk, R., Franks, C. & Dahlberg, J. Sorghum. In *Yield gains in major U.S. field crops* (eds Smith, S. et al.) 293–310 (Wiley, 2014). <https://doi.org/10.2135/assaspecpub33.c11>.
4. Kean, E. G., Ejeta, G., Hamaker, B. R. & Ferruzzi, M. G. Characterization of carotenoid pigments in mature and developing kernels of selected yellow-endosperm sorghum varieties. *J. Agric. Food Chem.* **55**, 2619–2626 (2007).
5. Fernandez, M. G. S. et al. Collection and characterization of yellow endosperm sorghums from West Africa for biofortification. *Genet. Resour. Crop Evol.* **56**, 991–1000 (2009).
6. Cardoso, L. M. et al. Tocochromanols and carotenoids in sorghum (*Sorghum bicolor* L.): Diversity and stability to the heat treatment. *Food Chem.* **172**, 900–908 (2015).
7. Cruet-Burgos, C. et al. Advancing provitamin A biofortification in sorghum: Genome-wide association studies of grain carotenoids in global germplasm. *Plant Genome* <https://doi.org/10.1002/tpg2.20013> (2020).
8. Owens, B. F. et al. Genome-wide association study and pathway-level analysis of kernel color in maize. *G3* <https://doi.org/10.1534/g3.119.400040> (2019).
9. Liang, X.-G. et al. Cob color, an indicator of grain dehydration and agronomic traits in maize hybrids. *Crop J.* **10**, 582–586 (2022).
10. Khamphan, P. et al. Effects of mass selection on husk and cob color in five purple field corn populations segregating for purple husks. *Agriculture* **10**, 311 (2020).
11. Rooney, W. L. Genetics and cytogenetics. In *Sorghum: Origin, History, Technology, and Production* (eds Smith, C. W. & Frederiksen, R. A.) (Wiley, 2000).

12. Rhodes, D. H. et al. Genome-wide association study of grain polyphenol concentrations in global sorghum [*Sorghum bicolor* (L.) Moench] germplasm. *J. Agric. Food Chem.* **62**, 10916–10927 (2014).
13. Fernandez, M. G. S. et al. Quantitative trait loci analysis of endosperm color and carotenoid content in sorghum grain. *Crop Sci.* **48**, 1732 (2008).
14. Chandler, K. et al. Genetic analysis of visually scored orange kernel color in maize. *Crop Sci.* **53**, 189–200 (2013).
15. Owens, B. F. et al. Genome-wide association study and pathway-level analysis of kernel color in maize. *G3 Genes Genomes Genetics* <https://doi.org/10.1534/g3.119.400040> (2019).
16. LaPorte, M.-F., Vachev, M., Fenn, M. & Diepenbrock, C. Simultaneous dissection of grain carotenoid levels and kernel color in biparental maize populations with yellow-to-orange grain. *G3 Genes Genomes Genetics* **12**, 5006 (2022).
17. Palmer, A. C. et al. Provitamin A carotenoid-biofortified maize consumption increases pupillary responsiveness among Zambian children in a randomized controlled trial. *J. Nutr.* **146**, 2551–2558 (2016).
18. Palmer, A. C. et al. Impact of biofortified maize consumption on serum carotenoid concentrations in Zambian children. *Eur. J. Clin. Nutr.* **72**, 301–303 (2018).
19. Burrow, G., Chopra, R., Hughes, H., Xin, Z. & Burke, J. Marker assisted selection in sorghum using KASP assay for the detection of single nucleotide polymorphism/insertion deletion. In *Sorghum: Methods and Protocols* (eds Zhao, Z.-Y. & Dahlberg, J.) 75–84 (Springer, 2019). https://doi.org/10.1007/978-1-4939-9039-9_6.
20. USDA ARS, G.-G. Germplasm Resource Information Network (GRIN). <https://npgsweb.ars-grin.gov/gringlobal/search>.
21. Casa, A. M. et al. Community resources and strategies for association mapping in sorghum. *Crop Sci.* **48**, 30 (2008).
22. Cruet-Burgos, C., Morris, G. P. & Rhodes, D. H. Characterization of grain carotenoids in global sorghum germplasm to guide genomics-assisted breeding strategies. *BMC Plant Biol.* **23**, 165 (2023).
23. Hu, Z., Olatoye, M. O., Marla, S. & Morris, G. P. An integrated genotyping-by-sequencing polymorphism map for over 10,000 sorghum genotypes. *Plant Genome* <https://doi.org/10.3835/plantgenome2018.06.0044> (2019).
24. Bradbury, P. J. et al. TASSEL: Software for association mapping of complex traits in diverse samples. *Bioinformatics* **23**, 2633–2635 (2007).
25. Lipka, A. E. et al. GAPIT: Genome association and prediction integrated tool. *Bioinformatics* **28**, 2397–2399 (2012).
26. Huang, M., Liu, X., Zhou, Y., Summers, R. M. & Zhang, Z. BLINK: A package for the next level of genome-wide association studies with both individuals and markers in the millions. *GigaScience* **8**, g154 (2019).
27. Morris, G. P. et al. Population genomic and genome-wide association studies of agroclimatic traits in sorghum. *Proc. Natl. Acad. Sci. U. S. A.* **110**, 453–458 (2013).
28. McCormick, R. F. et al. The Sorghum bicolor reference genome: Improved assembly, gene annotations, a transcriptome atlas, and signatures of genome organization. *Plant J.* **93**, 338–354 (2018).
29. R Core Team. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing (2022).
30. Mascher, M. et al. Genebank genomics bridges the gap between the conservation of crop diversity and plant breeding. *Nat. Genet.* **51**, 1076–1081 (2019).
31. Byrne, P. F. et al. Sustaining the future of plant breeding: The critical role of the USDA-ARS national plant germplasm system. *Crop Sci.* **58**, 451–468 (2018).
32. Cruet-Burgos, C. & Rhodes, D. H. Unraveling transcriptomics of sorghum grain carotenoids: A step forward for biofortification. *BMC Genom.* **24**, 233 (2023).
33. Cazzonelli, C. I. & Pogson, B. J. Source to sink: Regulation of carotenoid biosynthesis in plants. *Trends Plant Sci.* **15**, 266–274 (2010).
34. Uarrotta, V. G., Stefen, D. L. V., Leolato, L. S., Gindri, D. M. & Nerling, D. Revisiting carotenoids and their role in plant stress responses: From biosynthesis to plant signaling mechanisms during stress. In *Antioxidants and Antioxidant Enzymes in Higher Plants* (eds Gupta, D. K. et al.) 207–232 (Springer International Publishing, 2018). https://doi.org/10.1007/978-3-319-75088-0_10.
35. Meng, Y. et al. The MYB activator WHITE PETAL1 associates with MtTT8 and MtWD40-1 to regulate carotenoid-derived flower pigmentation in *Medicago truncatula*. *Plant Cell* **31**, 2751–2767 (2019).
36. Sagawa, J. M. et al. An R2R3-MYB transcription factor regulates carotenoid pigmentation in *Mimulus lewisii* flowers. *New Phytol.* **209**, 1049–1057 (2016).
37. Makanza, R. et al. High-throughput method for ear phenotyping and kernel weight estimation in maize using ear digital imaging. *Plant Methods* **14**, 49 (2018).
38. Yang, W. et al. Combining high-throughput phenotyping and genome-wide association studies to reveal natural genetic variation in rice. *Nat. Commun.* **5**, 5087 (2014).
39. Tanabata, T., Shibaya, T., Hori, K., Ebana, K. & Yano, M. SmartGrain: High-throughput phenotyping software for measuring seed shape through image analysis. *Plant Physiol.* **160**, 1871–1880 (2012).
40. ICRISAT (International Crops Research Institute for the Semi-Arid Tropics). Proceedings of the International Symposium on Sorghum Grain Quality, 28–31 October 1981, Patancheru, A.P., India. 407 (1982).
41. van Dongen, J. T. et al. Phloem import and storage metabolism are highly coordinated by the low oxygen concentrations within developing wheat seeds. *Plant Physiol.* **135**, 1809–1821 (2004).
42. Zheng, Y., Wang, Z. & Gu, Y. Development and function of caryopsis transport tissues in maize, sorghum and wheat. *Plant Cell Rep.* **33**, 1023–1031 (2014).
43. Harjes, C. E. et al. Natural genetic variation in lycopene epsilon cyclase tapped for maize biofortification. *Science* **319**, 330–333 (2008).
44. Saenz, E., Abdala, L. J., Borrás, L. & Gerde, J. A. Maize kernel color depends on the interaction between hardness and carotenoid concentration. *J. Cereal Sci.* **91**, 102901 (2020).
45. Woodruff, B. J., Cantrell, R. P., Axtell, J. D. & Butler, L. G. Inheritance of Tannin Quantity in Sorghum. *J. Hered.* **73**, 214–218 (1982).
46. Diepenbrock, C. H. et al. Eleven biosynthetic genes explain the majority of natural variation in carotenoid levels in maize grain. *Plant Cell* **33**, 882–900 (2021).
47. Zunjare, R. U. et al. Influence of rare alleles of β -carotene hydroxylase and lycopene epsilon cyclase genes on accumulation of provitamin A carotenoids in maize kernels. *Plant Breed.* **136**, 872–880 (2017).
48. Owens, B. F. et al. A foundation for provitamin A biofortification of maize: Genome-wide association and genomic prediction models of carotenoid levels. *Genetics* **198**, 1699–1716 (2014).
49. Sestili, F. et al. Provitamin A biofortification of durum wheat through a TILLING approach. *Int. J. Mol. Sci.* **20**, 5703 (2019).
50. Singh, J. et al. Marker-assisted pyramiding of lycopene- ϵ -cyclase, β -carotene hydroxylase1 and opaque2 genes for development of biofortified maize hybrids. *Sci. Rep.* **11**, 12642 (2021).
51. Watkins, J. L. & Pogson, B. J. Prospects for carotenoid biofortification targeting retention and catabolism. *Trends Plant Sci.* **25**, 501–512 (2020).
52. Lamb, A. et al. Bioenergy sorghum's deep roots: A key to sustainable biomass production on annual cropland. *GCB Bioenergy* **14**, 132–156 (2022).
53. Cruet-Burgos, C. et al. Advancing provitamin A biofortification in sorghum: Genome-wide association studies of grain carotenoids in global germplasm. *Plant Genome* **13**, e20013 (2020).

Acknowledgements

This work received financial support provided by Foundation for Food & Agriculture Research (FF-NIA20-000000036). This publication is made possible by the generous support of the American people through the United States Agency of International Development (USAID). The contents are the responsibility of the authors and do not necessarily reflect the views of USAID or the United States Government. Program activities are funded by the USAID through the Innovation Lab for Crop Improvement under Cooperative Agreement No. 7200AA-19LE-00005.

Author contributions

DR, SB, and GM designed the study. RM, LB, and DR collected and analyzed the data, and wrote the manuscript. SB and GM edited the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-024-75451-9>.

Correspondence and requests for materials should be addressed to D.H.R.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2024